Camera & ImageView App

Aim

To develop a simple Android application using Kotlin to open the camera, capture an image, and display it in an ImageView.

Software Requirements

- Android Studio
- Kotlin
- Android SDK

Step 1: Create a new Android Studio project

- **Language**: Kotlin
- **Template**: Empty Views Activity
- **Min SDK**: API 21 (or default)
- **File Name**: cameraapp

📄 Step 2: Layout (`activity_main.xml`)

Very basic UI:
➡️ One **ImageView**
➡️ One **Button**

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="16dp">

    <ImageView
        android:id="@+id/imageView"
        android:layout_width="300dp"
        android:layout_height="300dp"
        android:background="#DDDDDD"
        android:scaleType="centerCrop" />

    <Button
        android:id="@+id/btnCamera"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Open Camera"
        android:layout_marginTop="20dp" />

</LinearLayout>
```

Step 3: Kotlin code (`MainActivity.kt`)

This is the **entire logic**. Simple and direct.

```
package com.example.cameraapp

import android.Manifest
import android.app.Activity
import android.content.Intent
import android.content.pm.PackageManager
import android.graphics.Bitmap
import android.os.Bundle
import android.provider.MediaStore
import android.widget.Button
import android.widget.ImageView
import androidx.appcompat.app.AppCompatActivity
import androidx.core.app.ActivityCompat

class MainActivity : AppCompatActivity() {

    lateinit var imageView: ImageView
    lateinit var btnCamera: Button

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        imageView = findViewById(R.id.imageView)
        btnCamera = findViewById(R.id.btnCamera)

        btnCamera.setOnClickListener {
            ActivityCompat.requestPermissions(
                this,
                arrayOf(Manifest.permission.CAMERA),
                100
            )
        }
    }

    override fun onRequestPermissionsResult(
        requestCode: Int,
        permissions: Array<out String>,
        grantResults: IntArray
    ) {
        if (requestCode == 100 && grantResults[0] ==
PackageManager.PERMISSION_GRANTED) {
            startActivityForResult(
                Intent(MediaStore.ACTION_IMAGE_CAPTURE),
                100
            )
        }
    }

    override fun onActivityResult(
        requestCode: Int,
        resultCode: Int,
        data: Intent?
    ) {
        if (requestCode == 100 && resultCode == Activity.RESULT_OK) {
            val photo = data?.extras?.get("data") as Bitmap
```

```
            imageView.setImageBitmap(photo)
        }
    }
}
```

Step 4: Add Camera Permission (VERY IMPORTANT)

Open **AndroidManifest.xml** and add **this line ABOVE <application>**:

```
<uses-permission android:name="android.permission.CAMERA"/>
```

Output

- Tap **Open Camera**
- Camera opens 📸
- Take photo
- Photo appears in **ImageView**